

바이오메디컬 시스템반도체 융합설계 인력육성 센터

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바이오메디컬 시스템반도체 융합설계 인력육성 센터

2025 Chip Design Contest

Edge Detection Accelerator Based on Otsu Method

Graduate School of Electronic and Electrical Engineering,
Kyungpook National University

Joowan Kang, Insu Jeong, Seungjun Jo, Eunhae Lee, Taejun Kim, and Byungin Moon

Introduction

Edge detection constitutes a fundamental technique that significantly enhances the performance of image-based systems by extracting object boundaries. In order to generate an edge image, it is essential to select a threshold that distinguishes between edge and non-edge regions; the Otsu method is widely recognized as an effective approach for computing an optimal threshold that segments an image into two classes. In the present study, to facilitate an efficient FPGA implementation of the Otsu method, a histogram-based iterative computation scheme[1] was employed to determine the maximum between-class variance.

Otsu Method

The Otsu method is a binarization technique that automatically determines optimal thresholds by analyzing image histograms. This approach computes the between-class variance for all possible threshold candidates and selects the threshold value that maximizes the between-class variance as the optimal threshold.

Hardware Architecture of Otsu-based Edge Detector

Fig. 1 shows the overall architecture of the proposed edge detection accelerator. The system processes input images through sequential Gaussian and Laplacian modules, and then passes the results to the Proposed Otsu module. The Proposed Otsu module uses a histogram-based iterative computation method to determine the optimal threshold values from the processed image data. These computed optimal thresholds are subsequently transmitted to the Binarization module, which applies the thresholds to convert the filtered images into binary edge images. The histogram-based iterative computation implemented in the Proposed Otsu module significantly reduces memory requirements and computational complexity compared to conventional methods, thereby achieving efficient edge detection in terms of hardware resource utilization.

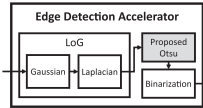


Fig. 1. Hardware architecture of the edge detection accelerator

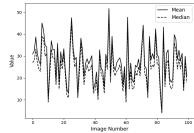


Fig. 2. Comparison of class mean difference calculations using mean and median

	Proposed Method	Logarithmic Approximation
Slice LUTs	9,442	19,184
Slice Registers	5,060	4,987
Accuracy of Binary Classification (%)	97.88	99.26

Table 1. Hardware resource utilization and accuracy of each method

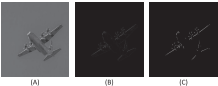


Fig. 3. Results of proposed method; (A) Original image, (B) LoG image, (C) Binary image

Conclusion

This study divides the input image into multiple blocks and calculates an independent Otsu threshold for each, dramatically reducing hardware resource usage. The results confirm that efficient edge detection can be achieved even on resource-constrained embedded platforms.

Reference

[1] J. Kang, I. Jeong, S. Jo, and B. Moon, "An Edge Detection Accelerator Using Otsu Method Based on Iterative Operation," The Transactions of the Korea Information Processing Society, 2025, vol. 14, no. 3, pp. 143-148.

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